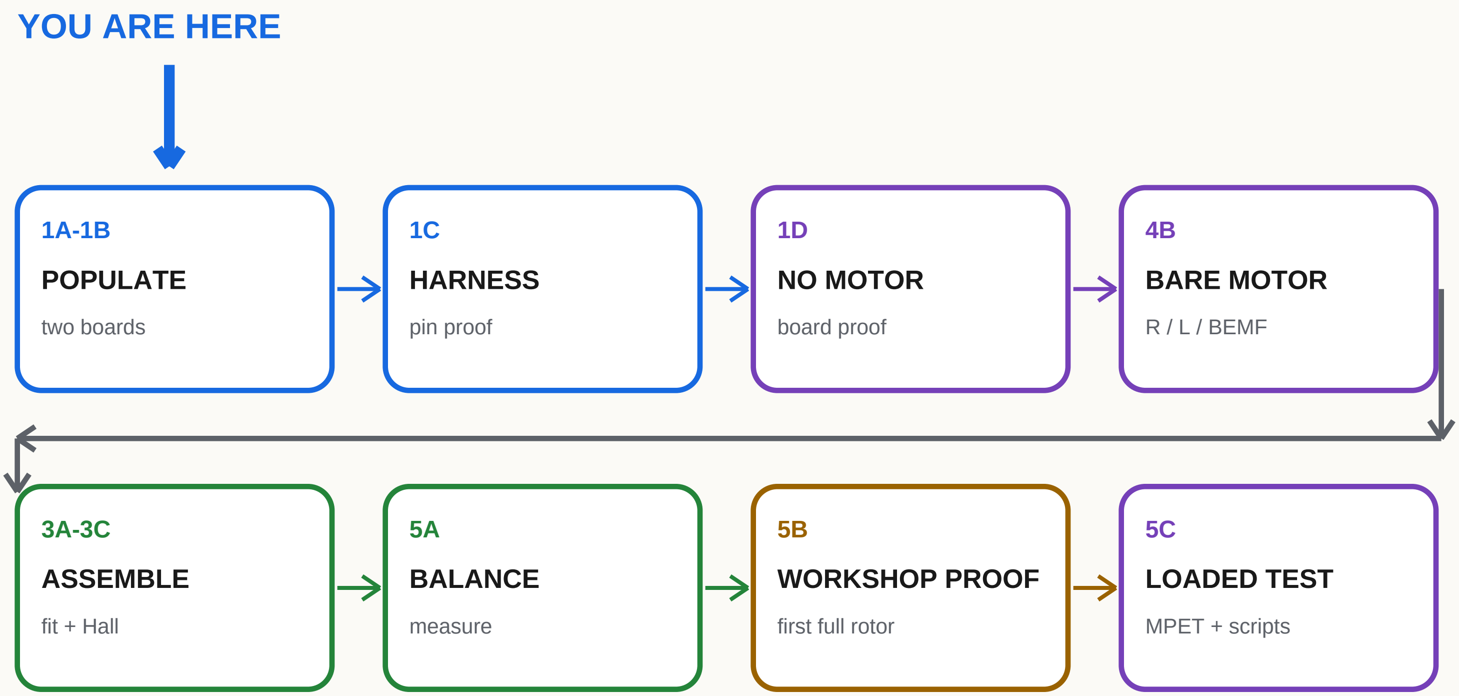
0A Active Integration Map

FIELD GUIDE



WHAT COUNTS AS DONE

- [ ] A result is measured and recorded, not just observed.
- [ ] Connector polarity and pin order are proved end-to-end.
- [ ] Installed unpowered checks precede gradual loaded work.
- [ ] A failed gate stops the next powered stage.
- [ ] After loaded MPET, verify config before start/speed qualification.
- [ ] Normal safety firmware + long USB J6; cutoff outside sweep.

BOUNDARIES KEPT OUT OF THIS ACTIVE BINDER

Michael performs installed work with one-step-at-a-time project guidance. Completed plate, tether, and catcher work stays closed. Deferred scope remains omitted unless explicitly reopened.

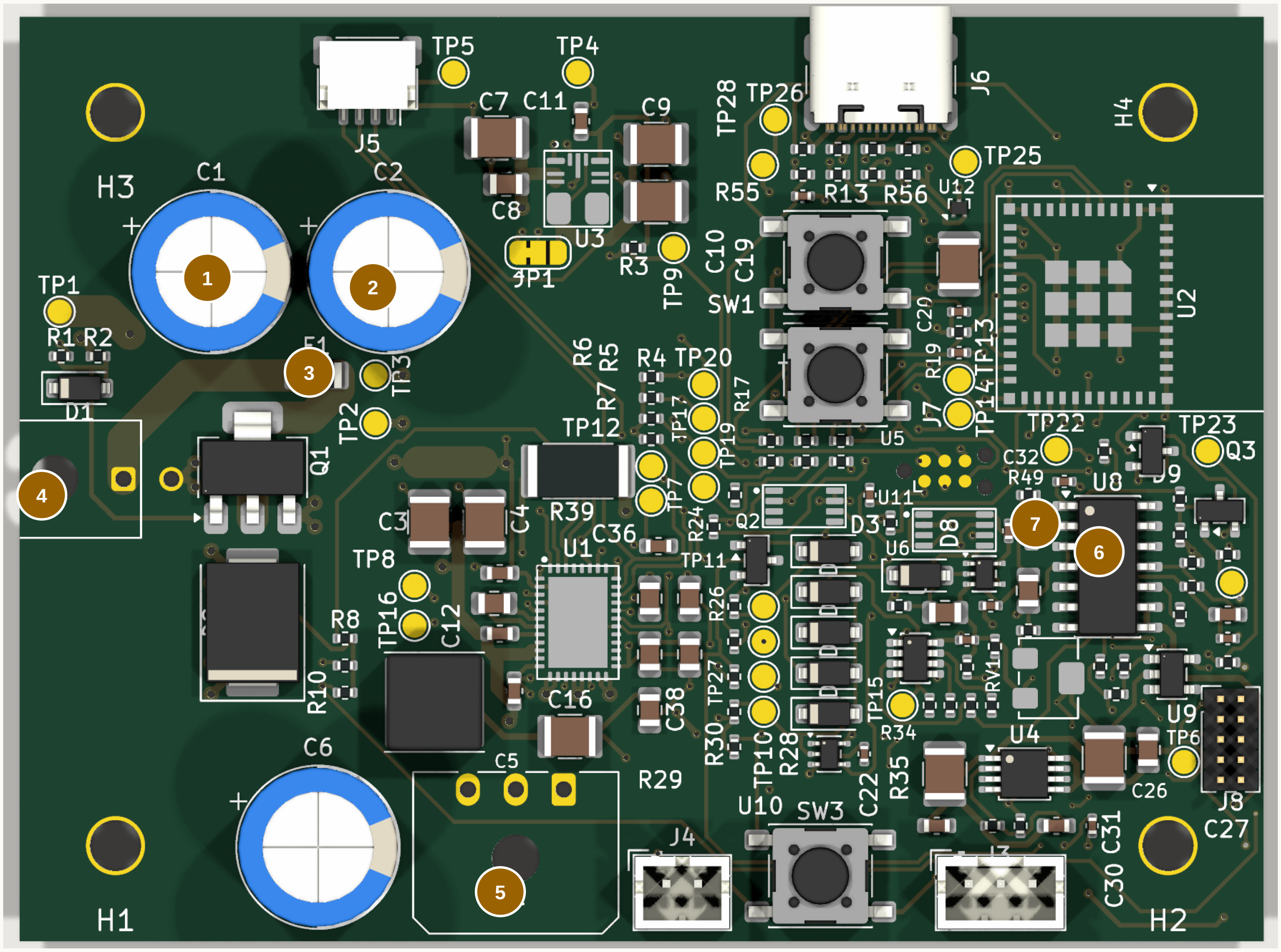
STOP / HANDOFF Start at 1A. Carry only the current sheet, board, and required tools to the bench.

SOURCE: docs/STATE.md; docs/integration.md; testing/test-matrix.csv

1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1	2 C2	3 F1	4 J1
5 J2	6 U8	7 C34	

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF C0G 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

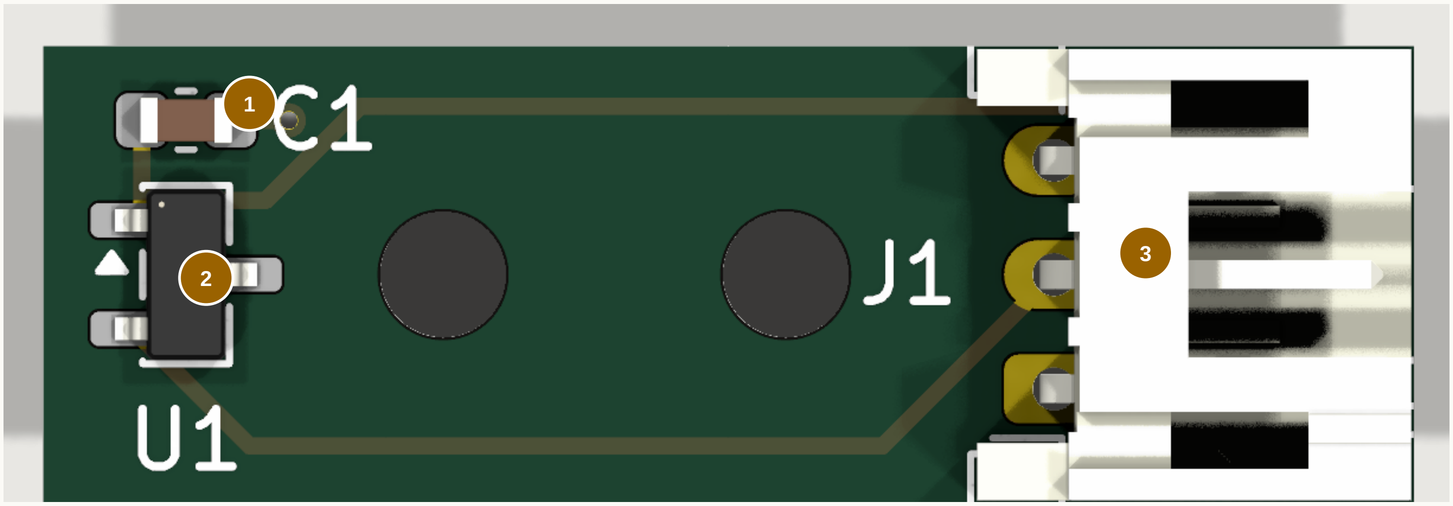
PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

SOURCE: pcb/pcb-01/pcb-01.kicad_pcb; bom/bom.csv; docs/electrical.md, SCH-06

1B PCB-02 Population + Soldering

FIELD GUIDE

TOP / MAGNET-FACING COMPONENT SIDE - KiCad target appearance



J1 / CABLE EXITS RIGHT →

1 C1
C0603C104K5RACTU, 100 nF 50 V X7R, 0603. Pin 1 / 3V3 is left; pin 2 / AGND is right.

2 U1
DRV5033FAQDBZR, SOT-23. Match package index to board triangle. Pin 1 upper-left = 3V3; pin 2 lower-left = HALL_TACH; pin 3 right = AGND.

3 J1
S3B-PH-K-S side-entry JST-PH. Top to bottom in this view: 3 AGND, 2 HALL_TACH, 1 3V3. Iron-solder last.

IRON METHOD: THIN SOLDER WIRE + FLUX

Flux + tin one pad → Place under microscope → Reheat to tack → Solder remaining pads → Inspect + wick bridges → Iron-solder J1

C1 - ONE-PAD TACK
Tin one pad lightly. Reheat it while placing C1, align the body, then solder the other end. Refresh the tack only if needed.

U1 - THREE LEADS
Flux, tack pin 3, correct alignment, then solder pins 1 and 2. Reflow the tack last if it needs a cleaner fillet.

J1 - LAST
Tack one pin, reheat while seating the body square and flush, finish the other pins, then revisit the tack.

FINAL INSPECTION

- ☐ All parts on magnet-facing side; sensor marked face will face magnet. Do not use magnet polarity as the orientation cue.
- ☐ Flux cleaned using its specified method; board completely dry.
- ☐ No short: 3V3 to HALL_TACH, 3V3 to AGND, or HALL_TACH to AGND.
- ☐ Microscope: wet fillets, no lifted lead, whisker, ball, or disturbed part; photograph board.

STOP / HANDOFF PCB-02 passes magnified inspection and unpowered three-net short checks. Do not power yet.

SOURCE: pcb/pcb-02/pcb-02.kicad_pcb; bom/bom.csv; docs/electrical.md, Hall daughterboard

1C Harness Build + Pin Proof

FIELD GUIDE

LOOK INTO THE MATING FACE; USE MOLDED PIN-1 MARKS, NOT WIRE COLOR MEMORY

POWER J1 - 18 AWG

1	2
RAW24	0V

Micro-Fit 2-way

DMM END-TO-END

MOTOR J2 - 18 AWG

1	2	3
W	V	U

Micro-Fit 3-way

HALL J3 / PCB-02 J1

1	2	3
3V3	TACH	AGND

PHR-3, straight-through

1-1 / 2-2 / 3-3

PROGRAM J7

1	2	3	4	5	6
3V3	TX	RX	EN	BOOT	GND

do not drive 3V3

EACH HARNESS: BUILD -> PROVE -> LABEL

Confirm housing view > Crimp + tug each contact > Insert to click > Continuity each pin > No cross-continuity > Label both ends

- [ ] Power: +24 V and 0 V preserved end-to-end.
- [ ] Hall: exactly 1-to-1, 2-to-2, 3-to-3.
- [ ] Motor: label positions W / V / U even if flying leads are not yet identified.
- [ ] Record length, gauge, wire colors, contact orientation, and result before sleeving.

J7 PIN 1 IS BOARD 3V3, NOT A POWER INPUT. Never let the programmer drive it while 24 V is connected.

STOP / HANDOFF Every harness is labeled, pull-tested, and signed off for pin continuity and no cross-continuity.

SOURCE: docs/electrical.md, SCH-07 connectors; testing/test-matrix.csv

TESTS: TACH-06

1D PCB-01 First Power, No Motor

FIELD GUIDE

WIRE THIS FIRST; PLACE PROBES BEFORE POWER

24 V BENCH SUPPLY
current limited

→

3 A FUSE + CUTOFF
within reach

→

PCB-01 J1
RAW24 / 0V

~~**J2 MOTOR**~~
~~DISCONNECTED~~

PROGRAMMER
TX / RX / EN / BOOT / GND; 3V3 crossed out

→

PROBES
DMM rails + scope VM with spring ground

NUMBERED TEST FLOW

1 RAILS
Brief power: 24 V, 3.3 V, AVDD, DVDD, DRVOFF; record current and heat.

2 HARDWARE CUTS
Force PGOOD, WDO, manual clear, OVERSPEED_N low one at a time -> DRVOFF rises without ESP help.

3 WATCHDOG
Static WDI: timeout 1.44-1.76 s, WDO pulse about 200 ms. 2 Hz falling edges prevent timeout.

4 STARTUP
Cold + slow-ramp power-up 20 times: U6 /PRE releases only after TACH_PGOOD_N.

5 RECOVERY
Remove/restore 24 V: disabled >=10 s; requires new command plus explicit arm.

6 TACH
Bridge disabled. Inject 0-3.3 V, 1-3% duty, settle >=30 s: reset ~3.00 Hz, trip ~3.33 Hz.

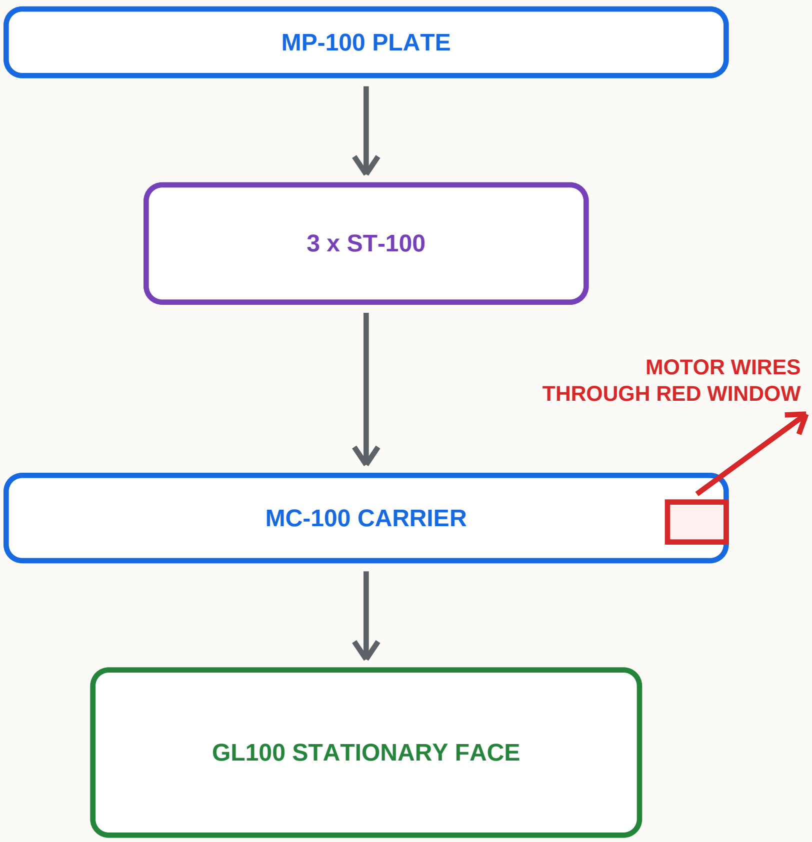
PASS LIMITS: VM <=35 V target; 40 V rejects. Persistent tach lock clears only on a genuine low-voltage power cycle. Any automatic re-arm blocks release.

STOP / HANDOFF All listed board/safety tests have recorded passes. Do not connect the motor after any failure.

3A Stationary Stack Dry-Fit

FIELD GUIDE

EXPLODED SIDE VIEW - NON-POWERED



MP-100 -> ST-100
3 x M6 x 16 flat-head from plate ceiling face; heads flush or below.

ST-100 -> MC-100
3 x M6 x 20 A4-80 + Nord-Lock pairs. Hand-start, square seating.

MC-100 -> GL100
4 x M4 x 12 A4-80 into the 60 mm stationary pattern. The 50 mm face rotates.

THREAD DEPTH
Target about 5.5 mm engagement; 6.0 mm motor-thread maximum. A bottoming screw is felt at hand torque: stop and correct.

RELEASE CHECK

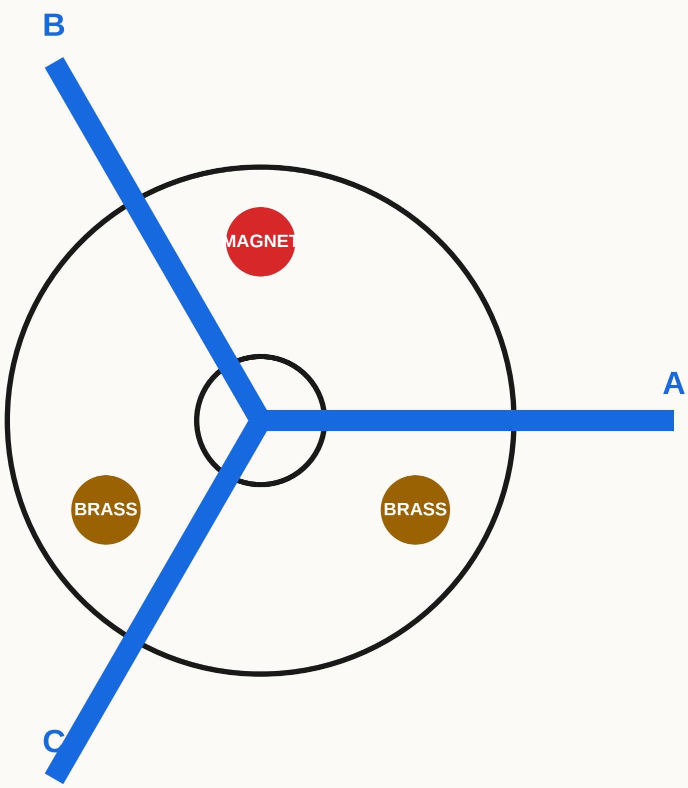
- ☐ All fasteners hand-start; no rocking or forced alignment.
- ☐ Owner-selected torque applied consistently; witness marks present.
- ☐ Stationary 60 mm and rotating 50 mm faces confirmed.
- ☐ Motor wire is unpinched and clear of sharp edges.
- ☐ Threadlocker only where specified; no screw bottoms.
- ☐ Stack remains non-powered until later gates pass.

STOP / HANDOFF Dry stack sits square without force, rocking, wire pinch, or bottoming screws.

SOURCE: docs/parts.md, GL100/ST-100/MC-100; bom/bom.csv

3B Rotor + Tach Inserts

FIELD GUIDE



HUB TO MOTOR

4 x M4 x 10 A4 flat-head from underside into 50 mm rotating pattern; heads 0.1-0.2 mm subflush. Pilot enters bore by hand only.

EACH BLADE

BP-100 v3 locating pins + captured M5 nuts; 4 x M5 x 22 per blade. Torque consistently and witness-mark.

THREE INSERT STATIONS

Epoxy retains one magnet + two brass slugs. Match complete insert-plus-epoxy station masses within 0.01 g after cure.

MASS + FINAL RECORD

MAGNET ______ g **BRASS B ______ g** **BRASS C ______ g** **MAX SPREAD ______ g**

- ☐ Pilot hand-fit; all four hub heads subflush.
- ☐ Epoxy fully cured; inserts cannot move.
- ☐ Critical fasteners torqued and witness-marked.
- ☐ All three blade stations have four correct bolts and captured nuts.
- ☐ Hand-rotate several revolutions: no rub, click, or changing clearance.
- ☐ Photograph rotor before balance measurements.

Do not use cad/BP-100.step for current fit; it is stale v2 geometry.

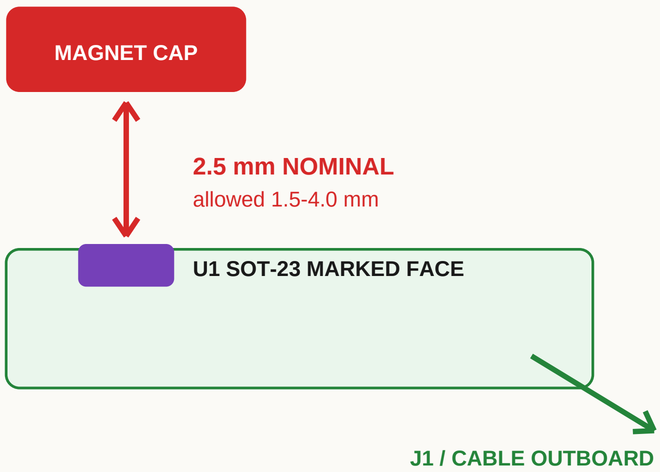
STOP / HANDOFF Complete rotor turns freely with correct retention; continue to 5A before powered motion.

SOURCE: docs/parts.md, RH-100/tach inserts; docs/blade-v2.md

3C Hall Gap + Electronics Bracket

FIELD GUIDE

SECTION THROUGH MAGNET AND SENSOR



COMMON CENTERLINE

Magnet and sensor element at radius 76.0 +/-0.5 mm.

PCB-02 MOUNTING

M2 holes on 6 mm pitch. At H1 use bare pan head or washer <=4.5 mm OD. Components face magnet.

BR-100

Strain-relieve Hall cable. Validate bracket around physical motor, board, magnet, and real gap.

PCB-01 BRACKET ENVELOPE

V1: 110 x 80 x 25 mm
V2: 120 x 90 x 30 mm SERVICE ENVELOPE

Four 6-8 mm M3 standoffs. Keep isolated mounting holes clear of circuit ground. Independent PCB retention and cable clamps. Preserve connector and cable-bend keepouts.

HAND-ROTATION RELEASE

- ☐ One clean pulse/revolution with magnet.
- ☐ All cables and connector bodies clear rotor.
- ☐ No false pulse with magnet removed.
- ☐ Gap remains in range through full hand rotation.

STOP / HANDOFF Hall pulse is clean; retained PCB and clamped cables clear all moving parts.

4A Flash + One Persistent CLI Session

FIELD GUIDE

BUILD / FLASH / TALK

```
cargo build --manifest-path firmware/Cargo.toml
```

```
espflash flash --port /dev/cu.usbmodem101 --non-interactive
  firmware/app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
firmware/target/debug/stillair --port /dev/cu.usbmodem101 state
```

BARE DEV-BOARD INPUT JUMPERS

GPIO22 -> 3V3
PGOOD good

GPIO21 -> 3V3
nFAULT idle high

GPIO14 -> GND
ALARM idle low

config stage → **Starting** → **15 s, no FG** → **NoRotation fault**

EXPECTED ON A BARE BOARD; IT DOES NOT GENERATE FG EDGES

SESSION RULES

ONE READER
Stop raw-log capture before espflash or CLI opens the port.

USE SCRIPT
Multi-step simulator work must stay in one session; separate invocations reset it.

CONFIG GATE
unverified = blocked; provisional = volatile bench; verified = release.

wait <state> --for <seconds> | wait speed <rpm> --within <rpm> --for <seconds>

stream <hz> --for <seconds> for CSV | config capture after complete measured configuration

Firmware refuses raw register/config writes unless stopped. Use controlled mpet run, not live raw commands. Simulator passes validate the harness, not the motor.

STOP / HANDOFF Firmware flashes; one persistent CLI session communicates and preserves scripted state.

SOURCE: AGENTS.md, Driving the fan; firmware/cli/src/main.rs

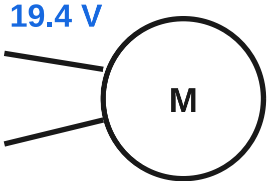
4B Unloaded First Spin

VOLATILE BENCH IMAGE

NO BLADES **Motor secured; reachable power cutoff; board and Hall tests passed**

1 POWER CYCLE

19.4 V



supply: 19.4 V / 2 A limit

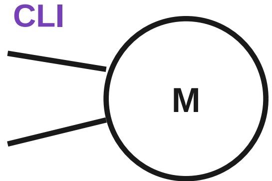
U-V: ________________

V-W: ________________

W-U: ________________

2 CONFIG STAGE

CLI



expect: provisional

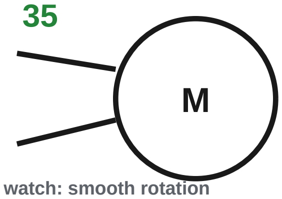
U-V: ________________

V-W: ________________

W-U: ________________

3 OBSERVED RUN

35



watch: smooth rotation

U-V: ________________

V-W: ________________

W-U: ________________

Double-align may make one or two small positioning ticks before smooth rotation.

GREEN LANE - DO NOW

- [ ] Power at 19.4 V with a 2 A supply limit; confirm normal idle draw.
- [ ] Run config stage after each motor-power cycle; require provisional.
- [ ] Issue a fresh run only after staging.
- [ ] Use script 03 at 35 RPM; watch continuously and keep the cutoff reachable.

LOADED-STAGE HANDOFF

- [ ] No unloaded MPET; script 02 requires the representative loaded rotor.
- [ ] The provisional image is for this unloaded first-spin test only.
- [ ] Never config apply the provisional image; it is volatile by design.
- [ ] After final apply and power cycle require config check: verified.

Factory-unverified is blocked: a normal run would invoke implicit MPET with zero gains. Stage first. VM <=35 V target; 40 V rejects.

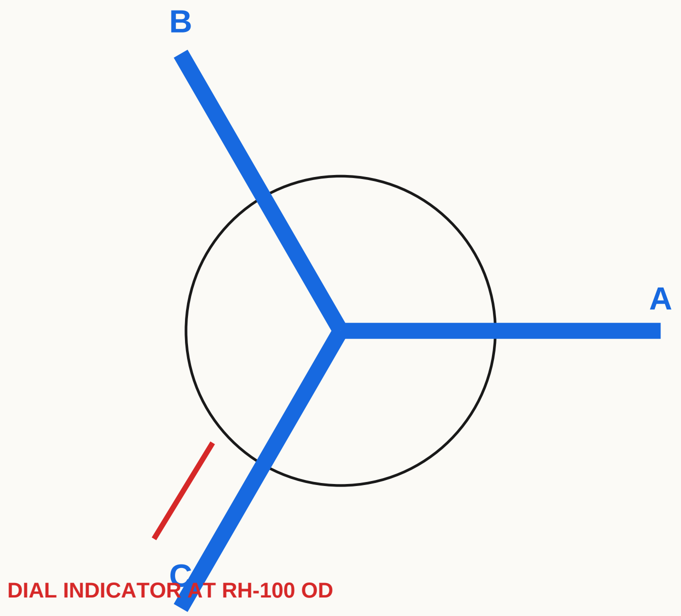
STOP / HANDOFF 35 RPM starts, turns smoothly, reports FG, and stops cleanly. Loaded tuning remains separate.

SOURCE: docs/controls.md, measured-data gate; docs/electrical.md

TESTS: DRV-01, PCB-02, CTL-08/09/10

5A Rotor Balance + Runout Record

FIELD GUIDE



HUB RUNOUT

<=0.10 mm TIR

BLADE TIP HEIGHT

all three in one indexed setup; spread <=0.5 mm

FIRST MOMENT

each blade; spread <=0.5%

A / B / C MEASUREMENT TABLE

STATION	MASS (g)	FIRST MOMENT	TIP (mm)	CORRECTION
A				
B				
C				

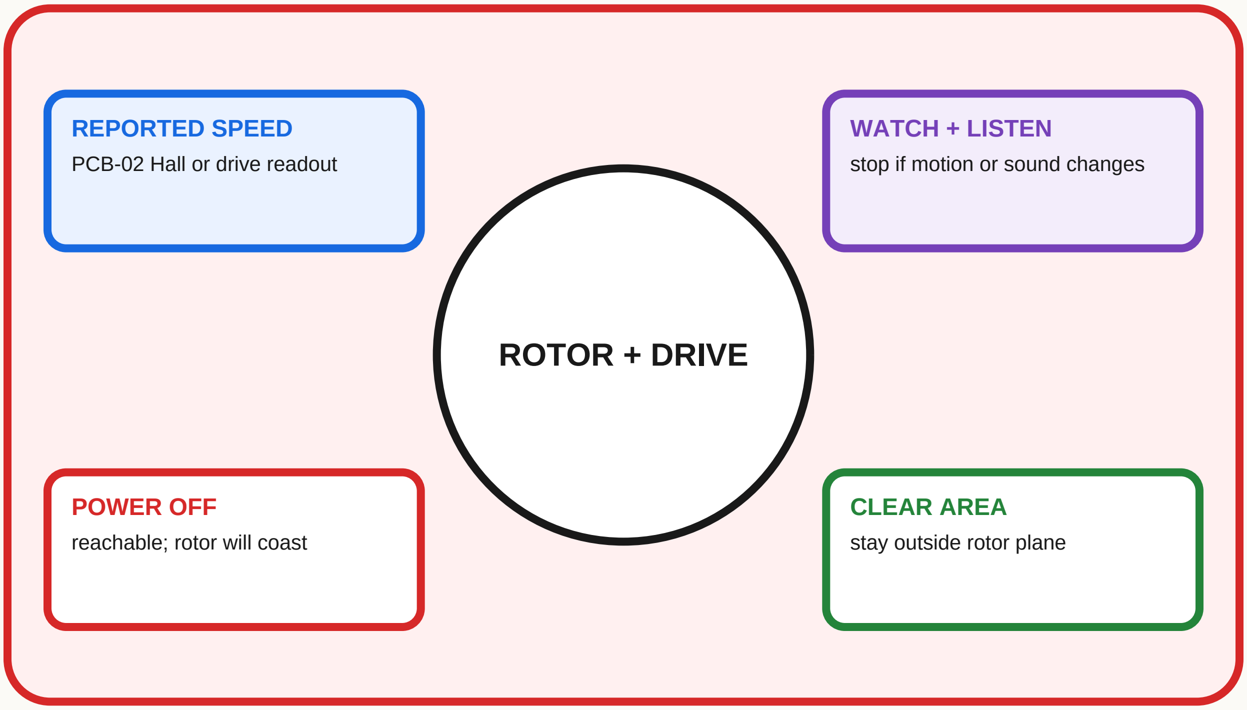
METHOD CHECK

- ☐ 0.01 mm indicator + 0.01 g calibrated scale.
- ☐ Correct only by documented slug method; remeasure.
- ☐ First powered check starts low; stop for wobble, vibration, rubbing, or unusual sound.
- ☐ Hand-clearance pass and witness-mark photo saved.

STOP / HANDOFF MEC-05 passes numerically; do not advance a rotor that only looks balanced.

5B Workshop Rotor Proof

FIELD GUIDE



GL100 PHASES DISCONNECTED FROM PCB-01 | NO SAFETY BYPASS

PCB-02 MAY REPORT HALL SPEED; ANALOG TRIP MAY LATCH BUT WILL NOT STOP THE EXTERNAL DRIVE

START LOW → 216 RPM → 2 min CW → STOP + INSPECT → 2 min CCW → STOP + INSPECT

270 RPM = CALCULATION ONLY x NEVER DYNAMICALLY TEST

BEFORE, DURING, AND RECORD

- ☐ Setup secured; rotor plane and nearby area clear.
- ☐ Start low; advance only while smooth and quiet.
- ☐ Balance, runout, clearance, witness marks recorded.
- ☐ Speed source responds and is recorded.
- ☐ Watch continuously; cutoff remains reachable.
- ☐ Inspect after each direction; stop on any change.

Speed source: ______________________________ Setup: ____

CW actual: ______ RPM result: __________ CCW actual: ______ RPM result: __________

Notes: ____

Date / initials: ____________________________________

STOP / HANDOFF Both directions pass with no abnormal motion, sound, contact, loosening, or damage.

SOURCE: testing/test-matrix.csv; docs/parts.md, design loads

5C Loaded Commissioning Script Card

HARDWARE VALUES PENDING

FINAL TEST CONNECTION

LAPTOP + CLI → LONG USB J6 → PCB-01 → MOTOR + ROTOR

NORMAL SAFETY FIRMWARE | 24 V VIA REACHABLE CUTOFF | CABLES OUTSIDE SWEEP

CONTROLLED MPET + GOLDEN IMAGE

- [ ] Run scripts/02-mpet-and-capture.txt only with the representative loaded rotor.
- [ ] Put only reviewed D-generation values into the golden IMAGE.
- [ ] Power-cycle after apply; config check must report verified.
- [ ] Review MTR_PARAMS, CURRENT_PI, and SPEED_PI against independent R / L / BEMF.
- [ ] While stopped, config apply performs one commit, waits 750 ms, polls self-clear, then verifies readback.
- [ ] If MPET faults or times out, firmware aborts it and revokes drive permission.

MPET result / image revision: ____

RUN THE NUMBERED FILES

02	`scripts/02-mpet-and-capture.txt`	loaded MPET + config capture
04	`scripts/04-loaded-speed-ladder.txt`	35 / 60 / 90 / 120 / 150 / 170 RPM ladder
05	`scripts/05-direction-check.txt`	low-speed direction check through verified stop
06	`scripts/06-observed-run.txt`	30-minute continuously observed run

WATCHED STOP RULES

- [ ] Keep the ordinary plug/cutoff reachable and watch continuously.
- [ ] Use reported FG diagnostics and your own observation; no external tachometer is required.
- [ ] Stop for visible wobble, increasing vibration, rubbing, or unusual sound.
- [ ] No permissive firmware or safety bypass. A stopped run may be repeated after inspection.

STOP / HANDOFF Loaded MPET, verified config, speed ladder, direction check, and observed run have recorded results.